

# STANDARD OPERATING PROCEDURE: SOP-704

## PRESSURE VESSEL RECERTIFICATION & MECHANICAL INTEGRITY MANDATE

Effective Date: 2026-01-01 | Authority: Sovereign Engineering Quality Directorate

### 3.1 Minimum Allowable Shell Wall Thickness

Under no operational conditions shall any pressure vessel operating above 50 PSI be recertified if any shell section exhibits a measured wall thickness of less than **12.0 mm**. Assets with thickness between 12.0 mm and 13.0 mm must undergo quarterly recalculation.

### 3.2 Hydrostatic Proof Testing Rules

Hydrostatic proof testing must be executed at exactly **1.5 times the Maximum Allowable Working Pressure (MAWP)**. For vessels rated at 100 PSI MAWP, the mandatory test pressure is **150.0 PSI** held for a minimum of 60 minutes with calibrated digital gauges.

### 4.0 Recertification Sign-Off & Physical Stamp Mandatory Requirement

Final recertification requires two criteria: (a) All mechanical inspection metrics meet or exceed Section 3 criteria, and (b) An authenticated **Secondary QA Stamp** is verified and recorded on the physical nameplate by a certified Level III NDT inspector. Automated or conditional approvals must withhold final return-to-service until the QA stamp is physically confirmed.